

The potential impact of reductions in international donor funding on tuberculosis in low-income and middle-income countries: a modelling study

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Summary

Background Tuberculosis programmes in many settings rely heavily on international donor funding. In 2025, the United States Agency for International Development (USAID) was dismantled, and other countries announced cuts to overseas development assistance. We quantified the potential epidemiological impacts on the tuberculosis burden attributable to these reductions in funding.

Methods We calibrated a deterministic tuberculosis model of *Mycobacterium tuberculosis* transmission, progression, and care to epidemiological indicators in selected low-income and middle-income countries. Calibration was done with the history matching with emulation method, implemented with the hmer package in R and the Approximate Bayesian computation Markov Chain Monte Carlo method. We projected three future scenarios with the following assumptions: that levels of funding in 2024 would continue, that USAID funding would be terminated from 2025, and that additional reductions in funding through The Global Fund to Fight AIDS, Tuberculosis and Malaria would occur (alongside termination of funding from USAID) in line with current donor announcements from 2025. We assumed a reduction in tuberculosis treatment initiation rates proportional to budget reductions for each scenario, estimating cumulative excess episodes of symptomatic tuberculosis and tuberculosis deaths for each scenario.

Findings We modelled 79 countries, representing 91% of global tuberculosis incidence and 90% of global tuberculosis mortality in 2023. Our modelling suggested that termination of USAID funding might lead to 1.4 million (95% uncertainty interval 1.1–1.7) excess tuberculosis episodes and 537 700 (451 900–662 300) excess deaths by 2035. Further reductions in funding in line with current announcements by the USA, France, the UK, and Germany could lead to 2.8 million (2.1–3.7), 257 600 (192 500–332 900), 206 000 (153 900–266 100), and 124 700 (93 200–161 000), additional episodes, respectively, of symptomatic tuberculosis and 1.0 million (0.8–1.3), 90 500 (72 400–112 800), 72 400 (57 900–90 100), and 43 800 (35 000–54 500) additional tuberculosis deaths, respectively, in the same period, relative to the scenario of termination of USAID funding.

Interpretation We estimate substantial potential impacts on tuberculosis morbidity and mortality due to reductions in international donor funding. Expanded support from domestic and international donors is essential to address immediate gaps in services for prevention, diagnosis, and treatment.

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Introduction

10.8 million people are estimated to have developed tuberculosis in 2023, with 1.25 million affected individuals estimated to have died in the same year.¹ Although tuberculosis remains the leading infectious cause of mortality globally, tremendous progress has been made to reduce morbidity and mortality. Between 2010 and 2020, estimated incidence decreased by 20% and mortality decreased by 40%.¹ This progress was interrupted by the COVID-19 pandemic, but recovery efforts have been strong, and the treatment coverage gap is now the smallest it has been since reporting began.¹ However, reaching WHO's ambitious End TB targets² of a 90% decline in incidence and a 95% decline in mortality between 2015 and 2035 would require large shifts in

current trends, for which funding remains a major challenge.³

Tuberculosis disproportionately affects low-income and middle-income countries (LMICs),⁴ and prevention, diagnostic, and treatment services in many settings rely heavily on international donor funding. The largest bilateral donor to the tuberculosis response has been the United States Agency for International Development (USAID), which contributed 19% of international donor funding reported by national tuberculosis programmes (NTPs) in 2022.³ However, The Global Fund to Fight AIDS, Tuberculosis and Malaria has been the primary source of international donor funding for tuberculosis since 2013, providing 66% of the international donor funding reported by NTPs in 2022. Although more than

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See [Comment](#) page 1490

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Research in context

Evidence before this study

Several modelling studies have begun to examine the long-term effects of cuts in funding from the USA. We searched PubMed and medRxiv for studies quantifying the effects of reductions in international donor funding for tuberculosis, published between Jan 1 and May 28, 2025, using search terms related to funding ("funding", "donor", "aid", or "assistance") and tuberculosis ("tuberculosis" and "TB"). We identified one preprint modelling potential impacts from the termination of United States Agency for International Development funding in 26 countries. To our knowledge, no studies have considered the further impact of expected reductions in overseas development assistance from other donor countries.

Added value of this study

Our modelling suggested that termination of USAID funding could lead to 537 700 excess tuberculosis deaths by 2035.

Further reductions in funding in line with current announcements by the USA, France, the UK, and Germany could lead to additional deaths from tuberculosis (1·0 million, 90 500, 72 400, and 43 800, respectively). The epidemiological impacts of further funding cuts would be greatest in some countries in Africa and in low-income countries.

Implications of all the available evidence

Evidence suggests that reductions in international donor funding have had substantial effects on service delivery, which will result in increased tuberculosis-associated morbidity and mortality. Expanded support from domestic and international donors is essential to address immediate gaps in prevention, diagnosis, and treatment, and more robust funding systems are needed going forward.

80 countries contribute to The Global Fund, the USA has been the largest donor to date. In The Global Fund's Seventh Replenishment (2023–25), the USA pledged 38% of the total amount pledged, with the next largest donors being France (10%), Germany (8%), and the UK (7%).⁵ Historic contributions to tuberculosis prevention and care by these agencies have saved the lives of millions of people with tuberculosis.⁶

In the first quarter of 2025, the USA abruptly dismantled USAID and cancelled funding for bilateral aid through the agency.⁷ At about the same time, other countries also announced reductions in contributions to bilateral aid programmes and multinational organisations. The UK, France, the Netherlands, and Belgium have announced cuts to official development assistance budgets ranging from 25% (in Belgium) to 40% (in the UK).⁸

USAID funding cuts have already interrupted prevention, diagnostic, and treatment services for tuberculosis,^{9–11} but the long-term effects of these disruptions have not been evaluated. We used mathematical modelling to estimate the epidemiological impacts attributable to the termination of USAID funding and the potential impact of future cuts in donor funding through The Global Fund in LMICs.

Methods

Model design and calibration

We used a dynamic, compartmental model of *Mycobacterium tuberculosis* transmission, progression, and care to simulate tuberculosis epidemic trajectories in selected LMICs. A detailed description of the model is in the appendix (pp 2–10).¹²

For each country in the study, we calibrated the model to eight epidemiological targets in 2023: country-specific tuberculosis incidence rates (for all ages, those aged 0–14 years, and those aged 15 years and older, separately), country-specific tuberculosis notification rates (for all

ages, those aged 0–14 years, and those aged 15 years and older, separately), country-specific tuberculosis mortality rate (for all ages), and the global proportion of asymptomatic tuberculosis episodes among infectious tuberculosis episodes (asymptomatic plus symptomatic; appendix pp 15–18). Models for countries classified as having a high tuberculosis burden due to HIV (appendix p 12) were fitted to at least three additional country-specific all-age targets: HIV prevalence and antiretroviral therapy coverage in 2021, and tuberculosis incidence rate in people living with HIV and tuberculosis mortality rate in people living with HIV in 2023 (appendix p 18).

Calibration was done with the history matching with emulation calibration method, implemented with the hmer package in R¹³ and the Approximate Bayesian computation Markov Chain Monte Carlo method, generating 200 fitted parameter sets per country. We used the distribution of results produced by these parameter sets to quantify parameter uncertainty.¹⁴ Further details of model simulation and calibration methods are in the appendix (pp 13–14).

Scenarios

For each country, we projected three future scenarios. First, we simulated a status quo scenario assuming that funding and programmatic activities would continue at 2024 levels. This scenario reflects international donor funding for tuberculosis over the last decade, which has remained relatively constant.³ Second, we modelled a scenario representing the effect of the termination of USAID funding to NTPs from 2025. Third, we simulated an additional reduction in funding through The Global Fund from 2025. For each donor that pledged at least 1% of the total pledged donations for The Global Fund's Seventh Replenishment (2023–25),⁵ we estimated the effect of changes in funding in line with announced

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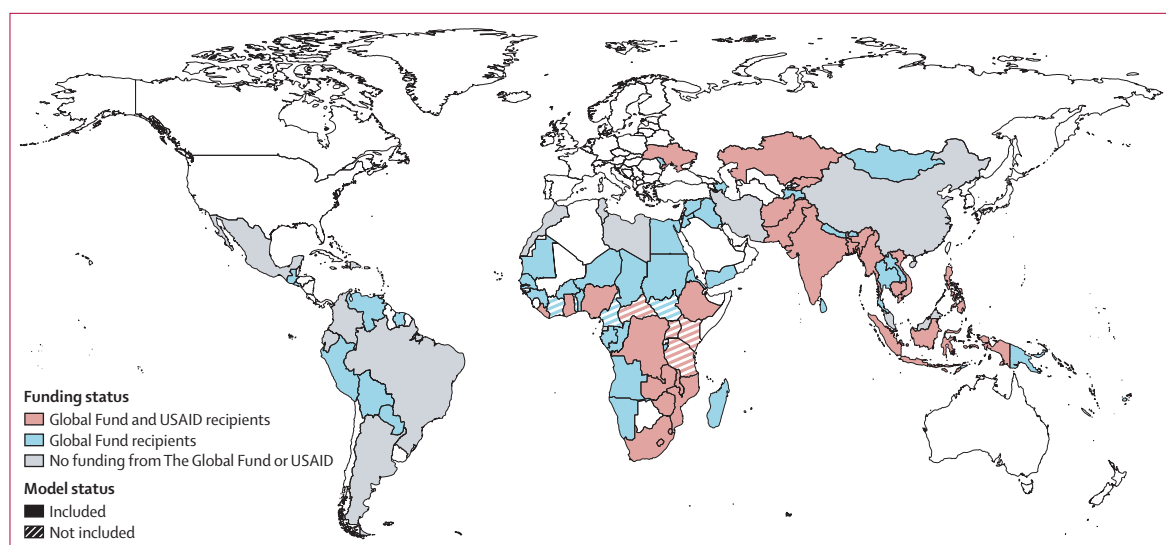


Figure 1: Countries included in the model and their funding status

Map shows countries that were modelled and received funding from USAID plus The Global Fund to Fight AIDS, Tuberculosis and Malaria (solid red), The Global Fund only (solid blue), or received no funding from either source (grey). Countries that were not modelled but received funding from USAID and The Global Fund (striped red) or The Global Fund only (striped blue) are also shown. Countries in white were not included. USAID=United States Agency for International Development.

changes in official development assistance.^{15–19} When considering the total effect of expected reductions in funding through The Global Fund, we included countries expected to increase their contributions. We also estimated the epidemiological impact of complete termination of funding to The Global Fund by donor. We assumed that all funding cuts were introduced in 2025, sustained into the future, and not replaced by other funding sources.

Funding cuts are likely to have a direct effect on tuberculosis treatment by limiting the accessibility of tuberculosis services and availability of tuberculosis diagnostics and treatment. Thus, we assumed a reduction in the rate of treatment initiation proportional to the budget reduction in each scenario. These reductions encompass decreased tuberculosis screening and active case finding, diagnosis (including rapid testing, drug susceptibility testing, specimen transport, and other diagnostic processes), contact investigation, community-supported tuberculosis care and stigma reduction, procurement and supply chains for diagnostics, external technical assistance, and capacity building and training.⁹ Budget reductions were calculated with budget data from WHO (appendix p 19),³ which has monitored funding for tuberculosis programmes since 2022. For each country, WHO data were used to calculate the proportion of total expected funding for all budget line items from USAID and from The Global Fund in 2023. We disaggregated The Global Fund contributions by donor using data on pledged donations for The Global Fund's Seventh Replenishment (2023–25).⁵

Epidemiological impact

For each scenario, we calculated the cumulative number of symptomatic tuberculosis episodes and tuberculosis

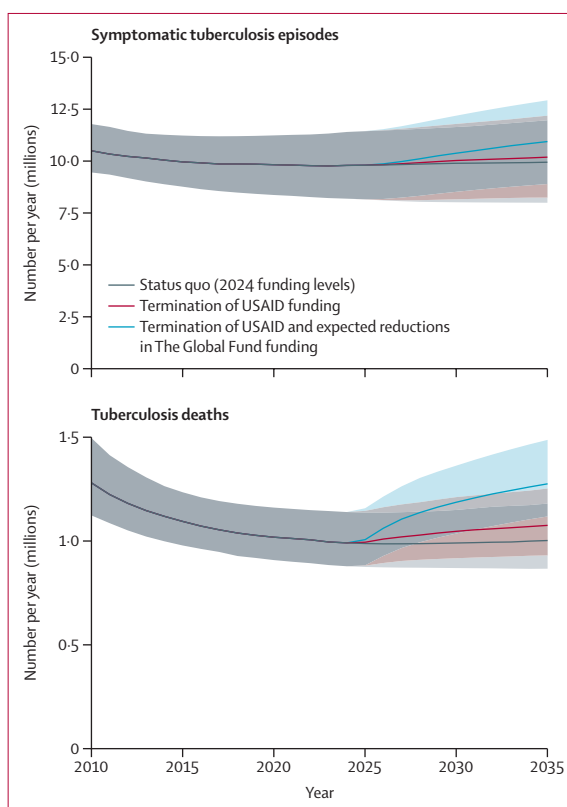


Figure 2: Symptomatic tuberculosis episodes and tuberculosis deaths (2010–35) projected to result from the status quo scenario and from termination of funding to national tuberculosis programmes from USAID, with and without announced reductions of funding to The Global Fund to Fight AIDS, Tuberculosis and Malaria across the 79 modelled countries Reductions in funding from The Global Fund included in the figure reflect announced reductions by donors that contribute at least 1% of total pledges. USAID=United States Agency for International Development.

deaths between 2025 and 2035 compared with the status quo. Results are presented as median values and 95% uncertainty intervals (UIs). We evaluated outcomes across all countries included and grouped results by WHO region and World Bank income group.

Role of the funding source

There was no funding source for this study.

Results

Of 111 LMICs with epidemiological and demographic data, complete budget data were available for 90.²⁰ Model calibrations were completed for 79 of these 90 countries (figure 1), representing 91% of global tuberculosis incidence and 90% of global tuberculosis mortality in 2023.¹ In 2023, 43 of the 79 countries received funding from The Global Fund, 23 received donor funding from both USAID and The Global Fund, and 13 did not receive funding from either source.²⁰ No countries received funding from USAID without also receiving funding from The Global Fund.

Of the 79 countries included in the analysis, the highest number of countries were from the WHO African region (n=29), followed by the WHO Region of the Americas (n=13), and the WHO Eastern Mediterranean region (n=12; appendix p 11). According to World Bank income groups, 34 countries were classified as lower-middle income countries, 29 as upper-middle income countries, and the remaining 16 as low-income countries.

Model-calibrated trends in tuberculosis incidence and mortality rates by WHO region are in the appendix (pp 20–21).

We estimated that termination of funding from USAID to NTPs might result in 1·4 million (UI 1·1–1·7) excess episodes of symptomatic tuberculosis and 537 700 (451 900–662 300) excess tuberculosis deaths by 2035, increases of 1·3% (1·0–1·7) and 5·0% (4·1–5·9), respectively, compared with the status quo scenario, which assumes that funding and programmatic activities continue at 2024 levels (figure 2, table 1).

The absolute and relative epidemiological impacts, although highly variable (dependent on country-specific reliance on international donor funding), were greatest in the WHO African region, with estimates of 761 400 (UI 531 500–964 000) excess episodes of symptomatic tuberculosis (an increase of 3·2% [UI 2·2–4·7] compared with the status quo scenario) and 338 900 (UI 270 800–441 900) excess tuberculosis-associated deaths by 2035 (an increase of 12·1% [UI 9·6–14·7] compared with the status quo scenario; table 2). The next greatest absolute epidemiological impact was in the WHO South-East Asia region, followed by the Western Pacific, the Eastern Mediterranean, and European regions. The WHO region of the Americas did not receive funding from USAID.

The absolute impact of termination of funding from USAID to NTPs on episodes of symptomatic tuberculosis was greatest in lower-middle-income countries, with 647 800 (UI 458 400–817 900) excess episodes predicted

	Excess episodes of symptomatic tuberculosis		Excess tuberculosis deaths	
	Median (95% UI)	% (95% UI)	Median (95% UI)	% (95% UI)
Termination of funding via USAID relative to the status quo scenario				
USA (–100%)	1 396 300 (1 095 700 to 1 720 000)	1·3% (1·0 to 1·7)	537 700 (451 900 to 662 300)	5·0% (4·1 to 5·9)
Expected reductions in funding via The Global Fund relative to termination of funding via USAID				
Total reductions	3 893 700 (2 890 400 to 5 085 600)	3·5% (2·6 to 5·1)	1 396 000 (1 114 200 to 1 748 100)	12·2% (9·2 to 14·6)
USA (–93%)	2 814 200 (2 096 100 to 3 663 000)	2·5% (1·9 to 3·7)	1 003 200 (801 700 to 1 256 500)	8·8% (6·6 to 10·4)
France (–37%)	257 600 (192 500 to 332 900)	0·2% (0·2 to 0·3)	90 500 (72 400 to 112 800)	0·8% (0·6 to 0·9)
Germany (–24%)	124 700 (93 200 to 161 000)	0·1% (0·1 to 0·2)	43 800 (35 000 to 54 500)	0·4% (0·3 to 0·4)
UK (–40%)	206 000 (153 900 to 266 100)	0·2% (0·1 to 0·3)	72 400 (57 900 to 90 100)	0·6% (0·5 to 0·7)
Japan (+17%)	–79 100 (–102 000 to –59 100)	–0·1% (–0·1 to –0·1)	–27 700 (–34 500 to –22 200)	–0·2% (–0·3 to –0·2)
Canada (–26%)	102 400 (76 500 to 132 200)	0·1% (0·1 to 0·1)	35 900 (28 700 to 44 800)	0·3% (0·2 to 0·4)
European Commission (–35%)	108 300 (80 900 to 139 800)	0·1% (0·1 to 0·1)	38 000 (30 400 to 47 300)	0·3% (0·3 to 0·4)
Sweden (–10%)	11 900 (8 900 to 15 300)	0·01% (0·01 to 0·02)	4 200 (3 300 to 5 200)	0·04% (0·03 to 0·04)
Norway (–12%)	10 000 (7 500 to 13 000)	0·01% (0·01 to 0·01)	3 500 (2 800 to 4 400)	0·03% (0·02 to 0·04)
Italy (+15%)	–11 900 (–15 400 to –8 900)	–0·01% (–0·02 to –0·01)	–4 200 (–5 200 to –3 300)	–0·04% (–0·04 to –0·03)
Netherlands (–29%)	22 500 (16 800 to 29 000)	0·02% (0·01 to 0·03)	7 900 (6 300 to 9 800)	0·1% (0·1 to 0·1)
Australia (–3%)	2 300 (1 700 to 3 000)	0·002% (0·002 to 0·003)	800 (600 to 1 000)	0·01% (0·01 to 0·01)

Values are presented as median numbers or percentages of excess symptomatic tuberculosis episodes or deaths, with 95% UIs representing the 2·5th and 97·5th percentile of the distribution of results. Countries are presented in order of size of donations to The Global Fund (percentages and 95% UIs are presented to two or three decimal places for smaller donors only). For termination of funding via USAID, percentages represent changes relative to status quo (2024 funding levels) scenarios; for expected changes in funding via The Global Fund, percentages represent changes relative to USAID scenarios. UI=uncertainty interval. USAID=United States Agency for International Development.

Table 1: Excess symptomatic tuberculosis episodes and tuberculosis deaths (2025–35) projected to result from termination in funding to national tuberculosis programmes from USAID and reductions in funding from The Global Fund to Fight AIDS, Tuberculosis and Malaria by donors that contribute at least 1% of total pledges

between 2025 and 2035 compared with the status quo scenario (table 2). However, the greatest relative impact was in low-income countries, with an increase of 4.7% (UI 2.6–7.1) compared with the status quo scenario. Similarly, the greatest absolute impact on excess tuberculosis-associated deaths was in lower-middle-income countries, with 252 000 (UI 188 700–336 200) excess tuberculosis deaths predicted, whereas the greatest relative impact was in low-income countries, with an increase of 18.1% (UI 13.8–23.3) compared with the status quo (table 2).

We estimated that reductions in funding through The Global Fund could further increase the number of symptomatic tuberculosis episodes and tuberculosis-associated deaths by 2035, over and above the increased numbers due to loss of funding from USAID to NTPs (table 1). Reductions in funding in line with current announcements by the USA, France, the UK, and Germany were predicted to lead to 2.8 million (UI 2.1–3.7), 257 600 (192 500–332 900), 206 000 (153 900–266 100), and 124 700 (93 200–161 000) additional episodes, respectively, of

symptomatic tuberculosis, relative to the scenarios of termination of funding via USAID, and 1.0 million (0.8–1.3), 90 500 (72 400–112 800), 72 400 (57 900–90 100), and 43 800 (35 000–54 500) additional tuberculosis deaths, respectively. Expected reductions in funding across all donors who pledge at least 1% of total pledges to The Global Fund might result in an additional 3.9 million (2.9–5.1) episodes of symptomatic tuberculosis and an additional 1.4 million (1.1–1.7) tuberculosis deaths, increases of 3.5% (UI 2.6–5.1) and 12.2% (9.2–14.6), respectively, relative to those predicted to be caused by termination of funding via USAID. The relative impact on episodes of symptomatic tuberculosis and tuberculosis deaths was estimated to be greatest in the WHO Eastern Mediterranean region, with increases of 8.7% (3.8–14.1) and 32.1% (22.0–42.4), respectively, and in low income countries, with increases of 6.2% (3.8–9.2) and 22.0% (17.7–27.4), respectively, relative to termination of funding via USAID.

Total termination of all funding through The Global Fund to NTPs might result in a further 11.6 million (UI 8.5–15.8) excess episodes of symptomatic

	Excess episodes of symptomatic tuberculosis		Excess tuberculosis deaths	
	Median (95% UI)	% (95% UI)	Median (95% UI)	% (95% UI)
Termination of funding via USAID relative to the status quo scenario				
WHO region				
African region	761 400 (531 500 to 964 000)	3.2% (2.2 to 4.7)	338 900 (270 800 to 441 900)	12.1% (9.6 to 14.7)
Region of the Americas	NA	NA	NA	NA
Eastern Mediterranean region	30 400 (15 700 to 46 300)	0.3% (0.1 to 0.5)	10 500 (6700 to 15 800)	1.4% (0.9 to 2.1)
European region	4200 (2400 to 6900)	0.6% (0.3 to 1.1)	1600 (–700 to 4200)	1.7% (–0.8 to 4.4)
South-East Asia region	446 300 (272 400 to 631 400)	0.8% (0.5 to 1.3)	160 900 (109 900 to 220 200)	2.7% (1.8 to 3.5)
Western Pacific region	155 600 (68 000 to 255 900)	0.9% (0.3 to 1.4)	27 400 (–1400 to 63 600)	2.9% (–0.1 to 6.1)
World Bank income group				
Low income	541 800 (338 600 to 746 500)	4.7% (2.6 to 7.1)	236 200 (175 900 to 324 000)	18.1% (13.8 to 23.3)
Lower-middle income	647 800 (458 400 to 817 900)	1.0% (0.6 to 1.3)	252 000 (188 700 to 336 200)	3.7% (2.8 to 4.6)
Upper-middle income	222 000 (90 000 to 343 700)	0.8% (0.3 to 1.3)	55 900 (16 200 to 85 400)	2.0% (0.6 to 3.2)
Expected reductions in funding via The Global Fund relative to termination of funding via USAID				
WHO region				
African region	1 053 700 (780 600 to 1 279 300)	4.2% (2.9 to 6.0)	497 300 (392 500 to 656 900)	15.6% (12.1 to 19.2)
Region of the Americas	8000 (5400 to 11 500)	0.3% (0.2 to 0.4)	2600 (–200 to 5000)	0.8% (–0.1 to 1.6)
Eastern Mediterranean region	822 000 (375 000 to 1 257 400)	8.7% (3.8 to 14.1)	250 500 (153 300 to 348 800)	32.1% (22.0 to 42.4)
European region	9800 (5500 to 16 000)	1.5% (0.7 to 2.3)	4300 (–1100 to 10 300)	4.5% (–1.3 to 10.8)
South-East Asia region	1 790 400 (917 400 to 2 854 500)	3.3% (1.7 to 5.5)	574 800 (300 300 to 837 500)	9.5% (4.5 to 13.3)
Western Pacific region	272 000 (166 100 to 385 800)	1.4% (0.8 to 2.2)	69 100 (21 500 to 121 000)	7.2% (2.3 to 11.0)
World Bank income group				
Low income	758 800 (509 500 to 991 400)	6.2% (3.8 to 9.2)	339 500 (263 300 to 434 600)	22.0% (17.7 to 27.4)
Lower-middle income	1 920 500 (1 362 900 to 2 404 500)	2.8% (1.9 to 3.9)	736 400 (597 100 to 901 000)	10.3% (8.0 to 13.0)
Upper-middle income	1 269 600 (490 500 to 2 168 800)	4.3% (1.7 to 7.8)	328 600 (72 200 to 548 500)	12.1% (2.7 to 19.9)

Values are presented as median numbers or percentages of excess symptomatic tuberculosis episodes or deaths, with 95% UIs representing the 2.5th and 97.5th percentile of the distribution of results. Donors contribute at least 1% of total pledges to The Global Fund and are listed by WHO region and World Bank income group. For termination of funding via USAID, percentages represent changes relative to status quo scenarios (2024 funding levels); for expected changes in funding via The Global Fund, percentages represent changes relative to the scenario with termination of funding from USAID. NA=not applicable. UI=uncertainty interval. USAID=United States Agency for International Development.

Table 2: Excess symptomatic tuberculosis episodes and tuberculosis deaths (2025–35) projected to result from termination in funding to national tuberculosis programmes from USAID and reductions in funding from The Global Fund to Fight AIDS, Tuberculosis and Malaria

tuberculosis and 4.4 million (3.3–5.5) excess tuberculosis deaths by 2035 (appendix p 23). These estimates are equivalent to increases of 10.3% (7.1–15.8) and 38.1% (27.0–47.8%), respectively, relative to projections from the scenario in which only funding from USAID is terminated. 1.1 million (0.9–1.4) deaths might result from termination of funding to The Global Fund from the USA, with 251 500 (201 300–314 200) deaths resulting from termination of funding from France, 186 900 (149 700–233 400) resulting from termination of funding from Germany, 184 300 (147 600–230 200) resulting from termination of funding from the UK, and 168 800 (135 100–210 700) resulting from termination of funding from Japan (appendix p 23).

Discussion

The sudden termination of USAID funding will reverse progress that has been made in recent years towards End TB targets,² erasing 12 years of progress in reducing tuberculosis incidence and a decade of progress towards reducing tuberculosis mortality by 2035. Just as the burden of tuberculosis is inequitably distributed, we project the cessation of USAID funding will have the largest epidemiological impact on those countries that have been most reliant on international donor funding, including many low-income countries and countries in the African region. Potential further reductions in funding due to reductions in official development assistance, modelled here through cuts in contributions to The Global Fund, are expected to further increase tuberculosis morbidity and mortality. These reductions would also disproportionately affect low-income countries and, in this analysis, countries in the Eastern Mediterranean region.

The epidemiological impact of reductions in international donor funding is inequitably distributed across countries, varying between and within income groups and geographical regions. We observed a relatively small proportional increase in symptomatic tuberculosis episodes and tuberculosis deaths attributable to termination of USAID funding compared with the status quo scenario (funding and programmatic activities continued at 2024 levels). This small increase is attributable to small USAID contributions to the NTPs of countries with the highest estimated numbers of incident tuberculosis cases in 2023 (3% of total NTP budget in India, 6% in Indonesia, 0% in China, 13% in the Philippines, and <1% in Pakistan). Although proportional increases in tuberculosis burden attributable to the elimination of USAID funding are low in these settings, greater epidemiological impacts are expected in other settings, such as the Democratic Republic of the Congo and Ethiopia, where USAID contributions accounted for more than a third of total NTP budgets. Donor funding through The Global Fund supports greater proportions of NTP budgets in many countries than does USAID, accounting for more than 90% of NTP budgets in 11 of the countries modelled,

including Viet Nam, Madagascar, and South Sudan, and more than half of NTP budgets in 29 of the countries modelled. Here, the epidemiological impacts are potentially larger than in countries that do not rely on The Global Fund for the majority of NTP funding. In addition, these local impacts have global implications for international tuberculosis targets, as well as cross-border consequences and repercussions for countries not directly affected.

The USA has been a leader in global health investment for decades and, through USAID and The Global Fund, has contributed nearly half of all international donor funding to NTPs.¹ These contributions have saved millions of lives.⁶ Our findings highlight the vulnerability of programmes that rely heavily on a single donor and support calls to diversify funding sources and reduce donor dependency.^{11,21–23} Even before recent reductions in international donor funding, support for the tuberculosis response fell far below targets set at the 2023 UN high-level meeting on tuberculosis.³ Thus, there is an urgent need to increase domestic funding for services for tuberculosis prevention, diagnosis, and treatment to build more robust systems.^{22,23} Immediate solutions to address the sudden departure of the USA from the funding landscape might require expanded support from other donors, including governments and philanthropic organisations, to address gaps in prevention, diagnosis, and treatment. However, in the longer term, real solutions require increased domestic investment by countries with high tuberculosis burdens, with an accompanying decreased reliance on external sources, to provide local ownership and identification of programmatic and research priorities.

The upcoming Eighth Replenishment of The Global Fund will be crucial. If the announced official development assistance cuts are reflected in reduced contributions to The Global Fund, we estimate further morbidity and mortality beyond that attributed to USAID cuts. It is essential that government donors fulfil their pledges to The Global Fund's 2023–25 funding cycle and uphold, or increase, their support in the Eighth Replenishment (2026–28) meeting in 2025. In addition, NTPs will need to identify approaches to achieve greater efficiencies within existing, reduced-funding envelopes. These efficiencies could include more integrated service delivery models, particularly with primary health-care and universal health coverage programmes. Additionally, prioritisation processes within tuberculosis programmes are needed to identify country-specific, optimised portfolios of interventions to enable the most appropriate use of remaining resources. At a global level, prioritisation is also required by The Global Fund and other stakeholders to identify countries most at risk and in need of any available additional support.

Our analysis had some limitations. We focused only on direct international donor funding to NTPs, emphasising

effects on diagnosis and treatment without quantifying effects on other aspects of tuberculosis prevention and care. We did not consider the epidemiological impact of reductions in funding to HIV or antiretroviral therapy programmes through USAID, the US President's Emergency Plan for AIDS Relief, or The Global Fund.^{24,25}

Given that HIV infection is the strongest risk factor for tuberculosis and that tuberculosis is the leading cause of death for people living with HIV, these funding cuts could have catastrophic effects, reversing the declines in HIV-associated tuberculosis seen over the past decades, particularly for some high-burden countries in the African region. Additionally, we have not considered the impact of the elimination of funding to support health systems, which might have indirect effects on tuberculosis programmes, or to multinational organisations that provide technical assistance and strategic support to affected countries. We also did not directly consider drug-resistant tuberculosis, which has greatly increased costs associated with both diagnosis and treatment and which might be more severely affected than drug-susceptible tuberculosis due to the reduced access to drug susceptibility testing and potential treatment interruptions associated with these funding cuts. Data were not available to quantify bilateral aid from countries other than the USA, so our analysis only considers contributions through The Global Fund for these countries. Although The Global Fund allocation policy means that reductions in funding to The Global Fund might not translate into proportional cuts in funding to tuberculosis programmes, we have made this simplifying assumption.

Our results rely on assumptions about the way changes in international donor funding affect tuberculosis prevention, diagnosis, and treatment and how these changes are implemented in the model. We assumed reductions in treatment initiation rates proportional to budget reductions covering areas such as tuberculosis screening and active case finding, diagnosis, contact investigation, and community-supported care; results from other modelling work highlight how different assumptions and implementation of those assumptions lead to different projections.²⁶ We have not modelled the effects of mitigation strategies that affected countries might introduce in response to reductions in funding, although we recognise that countries might implement emergency reallocation of domestic finances, enhanced bilateral aid, or changes in tuberculosis programme prioritisation. Additionally, we have only modelled scenarios in which funding reductions are sustained into the future and have not considered how alternative funding sources might mitigate epidemiological impacts. If some donors increase contributions beyond current expectations, we would expect reduced epidemiological impact relative to the results shown here.

We estimate substantial epidemiological impacts due to recent terminations and expected reductions in international donor funding that could reverse years of

progress to reduce tuberculosis morbidity and mortality. Expanded support from domestic and international donors is essential to address immediate gaps in prevention, diagnosis, and treatment of tuberculosis, and more robust funding systems are needed going forward.

Contributors

RAC: conceptualisation, software, formal analysis, visualisation, and writing—original draft. CFM: conceptualisation, methodology, and writing—review and editing. ASR: methodology, visualisation, and writing—review and editing. RB: software and writing—review and editing. TS: software and writing—review and editing. TOP-J: software and writing—review and editing. RMGJH: conceptualisation and writing—review and editing. RGW: conceptualisation, software, and writing—review and editing. KCH: conceptualisation, methodology, and writing—original draft. All authors had the opportunity to access and verify the data and were responsible for the decision to submit the manuscript for publication.

Declaration of interests

We declare no competing interests.

Data sharing

No individual-level participant data were used for this modelling study. The epidemiological data used are available from the WHO Global Tuberculosis Report and are summarised in the appendix (pp 15–18). Population estimates and projections are available from the UN Department of Economic and Social Affairs World Population Prospects 2022. The analytic code will be available immediately following publication on GitHub at <https://github.com/lshmt-tbmg/tbmod-pub-funding-impacts>, indefinitely, for anyone who wishes to access the data for any purpose.

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